

Question	Answer	Marks
9(a)	total power of radiation emitted (by the star)	B1
9(b)(i)	$F = L / (4\pi d^2)$	C1
	$= 9.86 \times 10^{27} / [4\pi \times (8.14 \times 10^{16})^2]$	A1
	$= 1.18 \times 10^{-7} \text{ W m}^{-2}$	
9(b)(ii)	$L = 4\pi\sigma r^2 T^4$	C1
	$9.86 \times 10^{27} = 4 \times \pi \times 5.67 \times 10^{-8} \times r^2 \times 9830^4$	
	radius = $1.22 \times 10^9 \text{ m}$	A1
9(c)	wavelength of peak intensity determined (from spectrum of star)	B1
	wavelength of peak intensity from object of known temperature determined	B1
	Wien's displacement law used or wavelength of peak intensity inversely proportional to temperature	B1

9702/43

Cambridge International AS & A Level – Mark Scheme

October/November 2022

PUBLISHED

Question	Answer	Marks
10(c)(i)	correctly predict when a (particular) nucleus will decay	D1